

Simultaneous Equations

Worksheet 1 (Non-Calculator)

1. W15/qp11/q23

A group of 15 adults and 12 children are going on a coach to a concert. The tickets for the coach cost \$a for each adult and \$c for each child. The total cost for the coach tickets is \$324.

- Show that $5a + 4c = 108$. [1]
- For a different group of 2 adults and 3 children the cost is \$53. Solve the simultaneous equations.

$$5a + 4c = 108$$

$$2a + 3c = 53$$

- Find the cost for a group of 4 adults and 5 children to travel on the coach. [1]

2. S16/qp12/q10

Solve the simultaneous equations.

$$6x + y = 1$$

$$4x - y = 4$$

3. W16/qp11/q18

Solve the simultaneous equations.

$$2x + 5y = 2$$

$$3x + 4y = -4$$

4. W16/qp12/q15

Solve the simultaneous equations.

$$3x + y = 9$$

$$2x + 3y = -8$$

5. S17/qp11/q11

Solve the simultaneous equations.

$$2x + 3y = 5$$

$$3x - y = -9$$

[3]

6. W17/qp11/q12

Solve the simultaneous equations.

$$2y = x + 12$$

$$3y = 11 - 2x$$

[3]

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Answer Key:

1.(a) $15a + 12c = 324$ (b) $a=16$ $c=799$ (c) 99

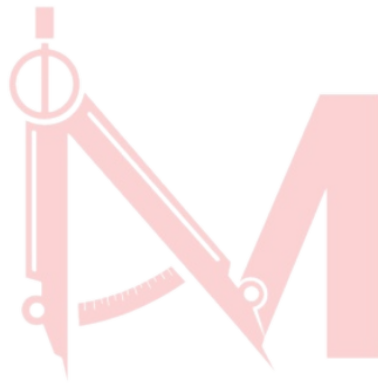
2. $x=0.5$ $y=-2$

3. $x=-4$ $y=2$

4. $x=5$ $y=-6$

5. $x=-2$ $y=3$

6. $x=-2$ $y=5$



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